1 Solution — GIAM 1.6 Problem 2

1.1 Problem

Base-n notation extends to real numbers by adding digits after a "decimal" point: e.g. $1.1011_2 = 1 \cdot 2^0 + 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} + 1 \cdot 2^{-4} = 1\frac{11}{16}$. Consider the binary number

$$x = .1010010001000010000010000001\ldots$$

Is this number rational or irrational? Why?

1.2 Answer

$$\boxed{x \text{ is } \textbf{irrational.}}$$

Why: its binary expansion never terminates and never becomes periodic — and a real number is rational *if and only if* its base-b expansion is eventually periodic. The runs of 0's between the 1's keep growing, so no repeating block can ever form.

1.3 Spotting the Pattern

Break the string at each 1:

$$x = .\underbrace{1}\,\underbrace{01}\,\underbrace{001}\,\underbrace{0001}\,\underbrace{00001}\,\underbrace{000001}\,\underbrace{0000001}\ldots$$

The k-th block is $(k-1)$ zeros followed by a 1, so the 1's land at positions

$$1,\ 3,\ 6,\ 10,\ 15,\ 21,\ 28,\ \ldots$$

after the point — the **triangular numbers** $T_k = \dfrac{k(k+1)}{2}$. Equivalently,

$$x = \sum_{k=1}^{\infty} 2^{-T_k} = 2^{-1} + 2^{-3} + 2^{-6} + 2^{-10} + 2^{-15} + \cdots$$

The gap (number of 0's) before the k-th one is $k-1$: the gaps run $1, 2, 3, 4, 5, 6, \ldots$ and **grow without bound**.

1.4 The Rationality Criterion

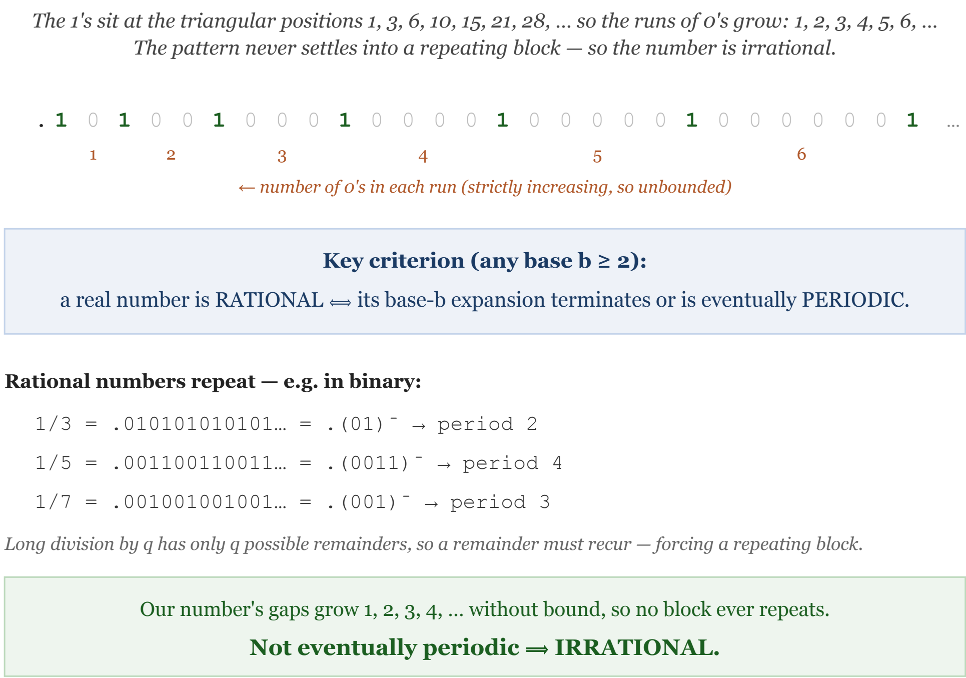

Figure 1.1: The binary digits with 1's highlighted at positions 1,3,6,10,15,21,28 — the triangular numbers — and the runs of zeros labelled 1,2,3,4,5,6, strictly increasing. The key criterion: a real number is rational iff its base-b expansion terminates or is eventually periodic. Rationals repeat: $1/3 = .(01)$ period 2, $1/5 = .(0011)$ period 4, $1/7 = .(001)$ period 3, because long division has only finitely many remainders. Our number's gaps grow without bound, so it is never periodic and therefore irrational.

The decisive fact — true in *every* base $b \geq 2$ — is:

$$\text{a real number is rational} \quad \Longleftrightarrow \quad \text{its base-}b\text{ expansion terminates or is eventually periodic.}$$

Our x does not terminate (there are infinitely many 1's). And it is **not eventually periodic**: a periodic tail would repeat some block of fixed length L forever, so the gaps between consecutive 1's would be bounded by L. But our gaps $1, 2, 3, \ldots$ exceed every bound. Hence no periodic tail exists, and x is irrational. ■

1.5 Remark — Why Rational $\Rightarrow$ Eventually Periodic

The "$\Rightarrow$" direction is just long division. To expand p/q in base b, repeatedly multiply the remainder by b and divide by q; each step's remainder lies in $\{0, 1, \ldots, q-1\}$ — only q possibilities. By the pigeonhole principle a remainder must **recur** within q steps, and once a remainder repeats the whole digit sequence repeats. So the period length is at most q. For example $1/3$ in binary cycles with period 2 ($.\overline{01}$), $1/7$ with period 3 ($.\overline{001}$), $1/5$ with period 4 ($.\overline{0011}$). A number whose digits refuse to settle into *any* such cycle therefore cannot be a fraction.

1.6 Remark — Why Eventually Periodic $\Rightarrow$ Rational

The converse is a geometric-series computation. An eventually periodic expansion is a terminating part plus a repeating block, and a pure repeating block of length L is a geometric series that sums to a fraction. In base ten this is the familiar $0.\overline{d_1 \cdots d_L} = \frac{d_1 \cdots d_L}{\underbrace{9 \cdots 9}_{L}}$; in base b the denominator is $b^L - 1$. For instance $.\overline{01}_2 = \frac{1}{2^2-1} = \frac{1}{3}$, matching $1/3$ above. Any eventually periodic string is thus a rational. Together with the previous remark this proves the biconditional the whole argument rests on.

1.7 Remark — The Same Number Written in Base Ten Is Still Irrational

Irrationality is a property of the *number*, not of the base you happen to write it in. If x were rational it would be eventually periodic in **every** base; being non-periodic in binary already settles it. Converting x to base ten would produce some other messy non-repeating string, but the conclusion is unchanged. This is worth stressing because the digits look base-specific — yet "rational or irrational" is base-independent.

1.8 Remark — A Cousin of Liouville's Number

This construction — placing 1's at a sparse, ever-thinning set of positions — is the classic recipe for building irrational (and even *transcendental*) numbers by hand. Liouville's famous constant $\sum_k 10^{-k!} = 0.110001000000000000000001000\ldots$ uses the factorials $k!$ (gaps growing even faster) and was the first number *proved* transcendental. Our $x = \sum_k 2^{-T_k}$ uses the triangular numbers; the growing gaps guarantee non-periodicity in exactly the same spirit. It stands in deliberate contrast to Problem 1: there we *approximated* irrationals like $\sqrt{2}$ by rationals, here we *manufacture* an irrational outright — and it neatly previews the section's headline result, the irrationality of $\sqrt{2}$, proved on the following page.